Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 26.15

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\begin{aligned}
&\int (4x^2+2x+1)(2x-1)^{-3}dx \\
&t = 2x-1 \left(\Rightarrow dt = 2dx \Rightarrow dx = \frac{dt}{2} \Rightarrow x = \frac{t+1}{2}\right) \\
&\int \left(4\left(\frac{t+1}{2}\right)^2 + 2\left(\frac{t+1}{2}\right)+1\right) t^{-3}\frac{dt}{2} \\
&= \int \left(\frac{4(t+1)^2}{4} + \frac{2(t+1)}{2} + 1\right) t^{-3}\frac{dt}{2} \\
&= \int \left((t+1)^2 + (t+1) + 1\right) t^{-3}\frac{dt}{2} \\
&= \int \left(t^2 + 2t + 1 + t + 1 + 1\right) t^{-3}\frac{dt}{2} \\
&= \int \left(t^2 + 3t + 3\right) t^{-3}\frac{dt}{2} \\
&= \frac{1}{2}\int \left(t^2 + 3t + 3\right) t^{-3}dt \\
&= \frac{1}{2}\int \left(t^{-1} + 3t^{-2} + 3t^{-3}\right) dt \\
&= \frac{1}{2}\left(\ln|t| - 3t^{-1} - \frac{3}{2}t^{-2}\right) + C \\
&= \frac{1}{2}\ln|t| - \frac{3}{2}t^{-1} - \frac{3}{4}t^{-2} + C \\
&= \frac{1}{2}\ln|2x-1| - \frac{3}{2(2x-1)} - \frac{3}{4(2x-1)^2} + C
\end{aligned}
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References